Implementation of Pose Estimation on SO(3) using the Passive Mahony Filter

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Submission Links

- GitHub repository: `https://github.com/Armaan-Raisinghani/dobs-assg3/tree/final`
- YouTube demonstration video: `https://youtu.be/ufHVakO_iEE`

1 Assignment Scope

This submission focuses on implementing the passive Mahony nonlinear complementary filter on $SO(3)$ using IMU and magnetometer data. The required supporting steps are:

1. Sensor interfacing (gyroscope, accelerometer, magnetometer), frame remapping to a common inertial convention (ENU), and calibration.
2. Gyro integration on $SO(3)$ using the exponential map / Rodrigues formula.
3. TRIAD implementation using accelerometer and magnetometer as a rotation measurement input.
4. Passive Mahony filter on $SO(3)$ with gyro-bias estimation.
5. Quaternion output and live 3D visualization for passive-filter validation.

2 Implementation Summary

2.1 Sensor Interfacing and ENU Alignment

All sensor readings were remapped into a consistent ENU-compatible body convention before fusion. In particular:

- Gyroscope channels were remapped to match the chosen body-axis definition used in the visualization.
- Accelerometer and magnetometer channels were remapped with sign and axis corrections so all modules used the same frame convention.
- Initial calibration was performed with the IMU body frame aligned with the chosen inertial reference.

2.2 Calibration

- Gyroscope: static bias estimated from averaged stationary samples.
- Accelerometer: static offset check under known gravity direction.
- Magnetometer: hard-iron and soft-iron compensation using calibration matrix and bias vector. This matrix was provided to us.

2.3 Gyro Integration on $SO(3)$

The attitude kinematics were integrated using

$$R_{k+1} = R_k \exp(\Omega_k^{\times} \Delta t),$$

with Rodrigues approximation for matrix exponential:

$$\exp(\theta K) = I + \sin(\theta) K + (1 - \cos\theta) K^2,$$

where $\theta = \|\Omega\| \Delta t$ and K is the unit-axis skew matrix. The practical procedure used in each loop was:

1. Read raw gyroscope channels and convert from deg/s to rad/s.
2. Apply ENU remapping and subtract the estimated static bias vector.
3. Build $\Omega^{\times}$ through the skew-symmetric operator.
4. Update R with the exponential-map step at fixed sample time Δt.
5. Reuse this rotation both for visualization and as the propagation model inside the passive filter pipeline.

2.4 TRIAD Measurement Block

TRIAD (provided baseline structure) was configured with accelerometer as the first vector and magnetometer as the second vector. This generated the rotation measurement used by the passive Mahony correction step. Given normalized body-frame vectors a_b and m_b, and normalized reference vectors a_r and m_r, TRIAD was computed using orthonormal bases:

$$q_b = a_b, \quad r_b = \frac{q_b \times m_b}{\|q_b \times m_b\|}, \quad s_b = q_b \times r_b,$$

$$q_r = a_r, \quad r_r = \frac{q_r \times m_r}{\|q_r \times m_r\|}, \quad s_r = q_r \times r_r.$$

Then

$$R_{\text{triad}} = [q_r \ r_r \ s_r][q_b \ r_b \ s_b]^{\top}.$$

The practical procedure was:

1. Apply accelerometer and magnetometer remapping to the common ENU-compatible convention.
2. Normalize both vectors to use directional information only.

3. Compute body/reference triads and construct R_{triad}.

4. Use R_{triad} as the instantaneous measurement matrix for passive-filter correction.

This block is sensitive to magnetic disturbance and linear acceleration, so it was treated as a correction/reference signal rather than the final estimate.

2.5 Passive Mahony Filter

The passive nonlinear complementary filter on $SO(3)$ code path (from provided base code) was configured and tuned with

- attitude correction from rotation error projection,
- integral gyro-bias estimation,
- proportional correction in the angular-rate update.

Quaternion streams were published for visualization in this order: **Gyro**, **TRIAD**, **Passive Mahony**. Gyro and TRIAD are included as intermediate/supporting references for debugging and validation. Using propagated estimate R_{est} and TRIAD measurement R_{triad}, the rotation error was formed as

$$R_E = R_{\text{est}}^\top R_{\text{triad}}, \quad \mathcal{P}_{\text{anti}} = \frac{1}{2}(R_E - R_E^\top),$$

and mapped to vector form

$$w = [\mathcal{P}_{\text{anti}}(3,2),\ \mathcal{P}_{\text{anti}}(1,3),\ \mathcal{P}_{\text{anti}}(2,1)]^\top.$$

Bias and corrected angular-rate updates were applied as:

$$\hat{b}_{k+1} = \hat{b}_k - k_i\,\Delta t\,w,$$

$$\Omega_{\text{corr}} = \Omega - \hat{b} + k_p\,w,$$

followed by $R_{\text{est},k+1} = R_{\text{est},k}\exp(\Omega_{\text{corr}}^\times \Delta t)$. The tuning workflow was: start with small k_i to avoid slow oscillatory bias drift, increase k_p until correction was responsive but not noisy, then iterate while checking heading stability and transient response in the live visualization.

3 Visualization Setup

The provided Python monitor (pygame), with label and stream mapping edits, displays three active cubes corresponding to:

1. Gyro integration output,
2. TRIAD output,
3. Passive Mahony output.

This supports side-by-side validation of the passive estimator against its input/reference blocks under identical motion.

4 Analysis and Observations

4.1 Gyro-only integration

- Without bias-correction, the gyro-only estimate starts drifting within seconds, due to the nature of the IMU bias and the integration process.
- Short-term motion is smooth and responsive, but long-term stability is poor.
- With bias correction, the reading is reliable only for up to 60 seconds, after which, even when left stable, there is anoticable drift of >30°s.
- Passive Mahony output is used as the reference estimate for comparison in the analysis.

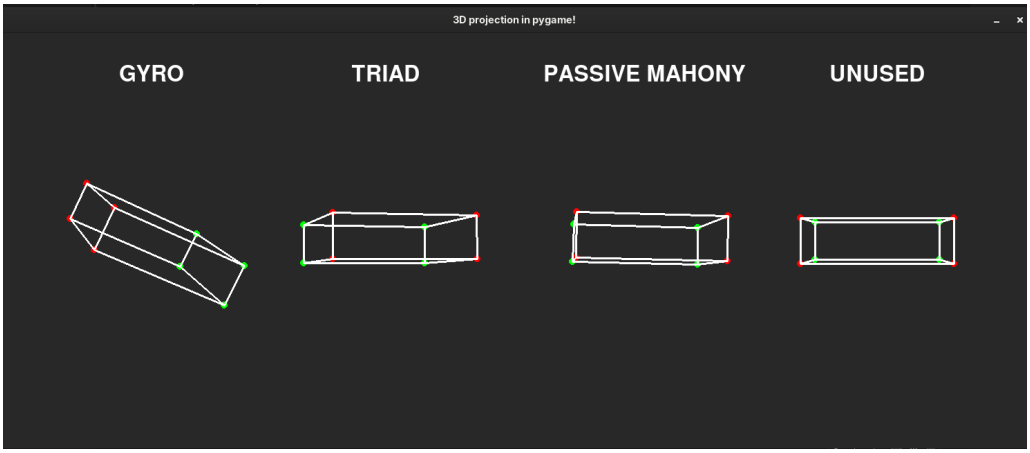

Figure 1: Gyro-only response, compared against Passive Mahony reference behavior.

4.2 TRIAD

The following behaviors were used to interpret passive-filter performance:

- TRIAD is always aligned with the true gravity and north vector, so it provides a true reference.
- However, it is noisy and can be unstable even at rest due to the noisiness of the accelerometer and magnetometer sensor readings.

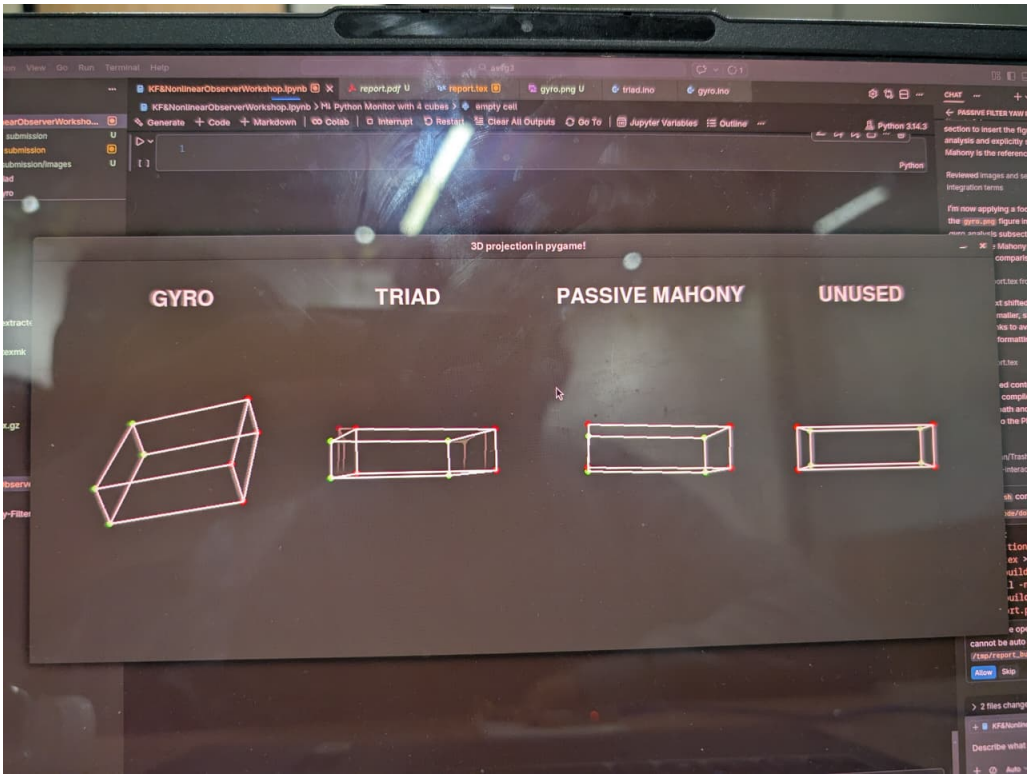

Figure 2: TRIAD output shown as a long-exposure shot taken from a camera, highlighting how shaky the estimate was.

4.3 Passive Mahony Filter

The following behaviors were used to interpret passive-filter performance:

- Passive mahony was by far the most accurate and stable estimate, with good transient response and heading stability.
- The passive filter was able to reject gyro drift and maintain a stable heading reference, even under moderate motion.
- However, it was not completely perfect. After strong shaking and some time at rest, it would still show some slow drift of $\approx 5^\circ$yaw, which is likely due to the limitations of the sensor calibration and the correction gains.

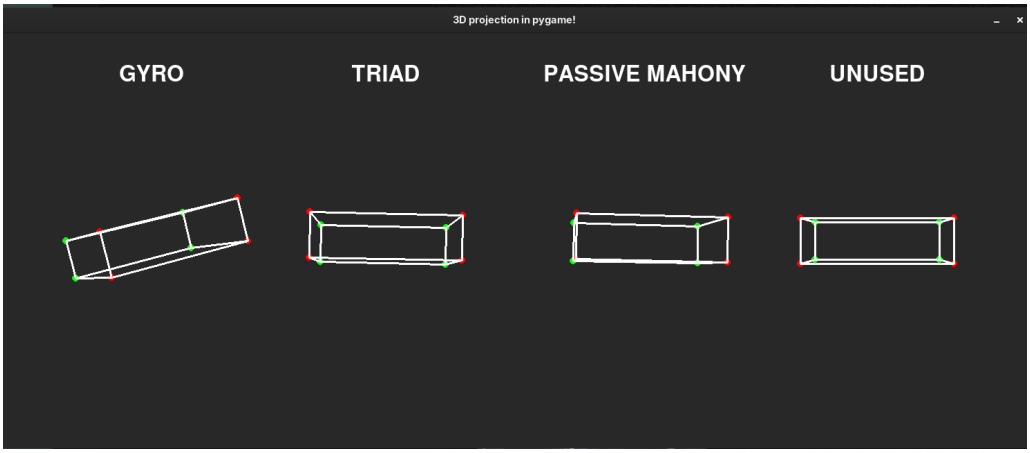

Figure 3: Passive Mahony filter output under the same motion sequence.

4.4 Final Conclusion

The passive Mahony filter was successfully implemented and it provided a significant improvement in attitude estimation stability and accuracy compared to gyro-only integration, while effectively using the TRIAD measurement as a correction reference. The live visualization was instrumental in tuning the filter parameters and understanding the behavior of each component under different motion conditions.

Appendix: Files Submitted

- Adapted Arduino sketches for gyro integration, TRIAD block, and passive Mahony pipeline.
- Provided Python visualization notebook with assignment-specific label/stream mapping updates.
- This report (PDF generated from `report.tex`).